

✓ SOP #4 – ATI (Analysis to Improve) for Chillers Incorporated

Goal: Identify long-term optimizations using data and performance trends to enhance chiller output, longevity, and energy savings

1. Key Optimization Metrics

Metric	Ideal Range or Target	Optimization Action
Energy Efficiency Ratio (EER)	>11 (Air Cooled), >15 (Water Cooled)	Add VFDs, improve heat transfer surfaces
Coefficient of Performance (COP)	>3.0	Monitor vs. historical load data
Delta-T (Evap. Side)	10-12°F	Improve balancing valve or flow controls
kW/Ton	<0.75 (high-efficiency)	Reduce oversizing, balance load zones
Runtime Consistency	≥80% during high-load season	Enable chilled water resets

2. Strategic Improvements

Add Heat Recovery: Reuse condenser waste heat for reheat or boiler support
 Reset Setpoints by OAT or Load: Reduce chilled water temp on milder days
 Install Flow Sensors & Remote Alerts: For real-time monitoring of surges or fouling
 Use Variable-Speed Drives: On condenser water pumps and fans
 Log runtime & efficiency weekly: Use CTS logs for trend tracking

3. Legal, Safety & Compliance Considerations

ASHRAE 90.1 & 189.1: Efficiency and energy requirements for chilled systems
 EPA Clean Air Act Section 608: Refrigerant handling, leaks, and records
 OSHA Standards: Safe access during tube pulling or refrigerant charge
 NFPA 70 (NEC): Electrical service to compressors and panels